

MCQ TEST - CO2 AT1

Total points 49/50

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✓ Which activation function suffers from vanishing gradients? * 1/1

- ☒ A) Sigmoid ✓
- ☐ B) ReLU
- ☐ C) Leaky ReLU
- ☐ D) ELU

✓ Which model is commonly used for feature extraction? * 1/1

- ☐ A) Decision Tree
- ☒ B) Autoencoder ✓
- ☐ C) Linear Regression
- ☐ D) Naïve Bayes



✓ ELU stands for: *

1/1

- ☒ A) Exponential Linear Unit ✓
- ☐ B) Extreme Learning Unit
- ☐ C) Elastic Linear Unit
- ☐ D) Efficient Learning Unit

✓ Which activation function outputs values between -1 and 1? *

1/1

- ☐ A) ReLU
- ☒ B) Tanh ✓
- ☐ C) Softmax
- ☐ D) ELU

✓ Softmax converts outputs into: *

1/1

- ☐ A) Binary values
- ☐ B) Feature vectors
- ☒ C) Probability distribution ✓
- ☐ D) Random numbers



✓ Which application uses Deep Learning to understand human language? * 1/1

- ☒ A) NLP ✓
- ☐ B) Compiler
- ☐ C) DBMS
- ☐ D) Operating System

✓ Self-driving cars use Deep Learning mainly for: * 1/1

- ☐ A) Memory allocation
- ☒ B) Object detection and decision making ✓
- ☐ C) Database indexing
- ☐ D) Encryption

✓ In Deep Learning, representations become: * 1/1

- ☐ A) Less abstract
- ☒ B) More meaningful and abstract ✓
- ☐ C) Random
- ☐ D) Unusable



✓ Which activation function is computationally the simplest? *

1/1

- ☐ A) Sigmoid
- ☐ B) Tanh
- ☒ C) ReLU
- ☐ D) Softmax



✓ Which activation function is zero-centered? *

1/1

- ☐ A) Sigmoid
- ☒ B) Tanh
- ☐ C) ReLU
- ☐ D) Softmax



✓ Which application commonly uses Autoencoders? *

1/1

- ☒ A) Noise removal
- ☐ B) Sorting
- ☐ C) Searching
- ☐ D) Routing



✓ The depth of a neural network refers to: *

1/1

- ☐ A) Number of neurons
- ☒ B) Number of hidden layers ✓
- ☐ C) Number of outputs
- ☐ D) Number of datasets

✓ Increasing the depth of a neural network mainly helps: *

1/1

- ☐ A) Reduce storage
- ☒ B) Learn complex hierarchical features ✓
- ☐ C) Reduce dataset size
- ☐ D) Increase randomness

✓ RBMs are mainly used for: *

1/1

- ☐ A) Classification only
- ☒ B) Feature learning ✓
- ☐ C) Sorting
- ☐ D) Searching



✓ Representation Learning refers to: *

1/1

- ☐ A) Manual feature extraction
- ☒ B) Learning useful features automatically from raw data ✓
- ☐ C) Data compression only
- ☐ D) Data visualization

✓ Which activation function is generally preferred in hidden layers of modern neural networks?

*1/1

- ☐ A) Sigmoid
- ☒ B) ReLU ✓
- ☐ C) Softmax
- ☐ D) Linear

✓ Which activation function is mainly used in multi-class classification? *

1/1

- ☐ A) Sigmoid
- ☐ B) ReLU
- ☒ C) Softmax ✓
- ☐ D) Tanh



✗ Which is NOT an example of Representation Learning? *

0/1

- ☐ A) Autoencoder
- ☐ B) Deep Neural Network
- ☐ C) Principal Component Analysis
- ☒ D) Manual Feature Engineering ✗

Correct answer

- ☒ B) Deep Neural Network

✓ Deep Learning is primarily based on: *

1/1

- ☐ A) Decision Trees
- ☒ B) Artificial Neural Networks ✓
- ☐ C) Random Forests
- ☐ D) K-Means Clustering

✓ Which model generally requires a larger amount of data? *

1/1

- ☐ A) Linear Regression
- ☐ B) Decision Tree
- ☒ C) Deep Learning ✓
- ☐ D) Naïve Bayes



✓ In an RBM, neurons within the same layer: *

1/1

- ☐ A) Are fully connected
- ☒ B) Are not connected
- ☐ C) Form cycles
- ☐ D) Use convolution



✓ Which of the following is an advantage of Deep Learning over traditional Machine Learning? *1/1

- ☐ A) Requires no data
- ☒ B) Performs automatic feature extraction
- ☐ C) Always faster to train
- ☐ D) Needs less computational power



✓ Face recognition systems mainly use: *

1/1

- ☒ A) Deep Learning
- ☐ B) Bubble Sort
- ☐ C) Binary Search
- ☐ D) AVL Trees



✓ Which type of Autoencoder removes noise from images? *

1/1

- ☐ A) Sparse Autoencoder
- ☒ B) Denoising Autoencoder ✓
- ☐ C) Variational Autoencoder
- ☐ D) Linear Autoencoder

✓ Which technique usually requires manual feature engineering? *

1/1

- ☐ A) Deep Learning
- ☐ B) Representation Learning
- ☒ C) Traditional Machine Learning ✓
- ☐ D) Reinforcement Learning

✓ The decoder reconstructs: *

1/1

- ☐ A) Labels
- ☒ B) Original Input ✓
- ☐ C) Hidden Layer
- ☐ D) Gradients



✓ Unsupervised learning uses: *

1/1

- ☐ A) Labeled data
- ☒ B) Unlabeled data ✓
- ☐ C) Test data only
- ☐ D) Validation data only

✓ An Autoencoder has: *

1/1

- ☐ A) Input layer only
- ☒ B) Encoder and Decoder ✓
- ☐ C) Hidden layer only
- ☐ D) Output layer only

✓ Restricted Boltzmann Machine (RBM) is: *

1/1

- ☐ A) Supervised model
- ☐ B) Reinforcement Learning model
- ☒ C) Unsupervised neural network ✓
- ☐ D) Decision Tree



✓ Which activation function outputs values between 0 and 1? *

1/1

- ☐ A) ReLU
- ☐ B) ELU
- ☒ C) Sigmoid
- ☐ D) Tanh



✓ Which of the following is NOT a Deep Learning application? *

1/1

- ☐ A) Machine Translation
- ☐ B) Fraud Detection
- ☐ C) Image Classification
- ☒ D) Bubble Sort



✓ What is the output of ReLU when $x = -5$? *

1/1

- ☐ A) -5
- ☐ B) 5
- ☒ C) 0
- ☐ D) 1



✓ Unsupervised neural networks mainly aim to: *

1/1

- ☐ A) Predict labels
- ☒ B) Discover hidden patterns ✓
- ☐ C) Increase hardware speed
- ☐ D) Encrypt data

✓ The formula for Sigmoid activation is: *

1/1

- ☐ A) $\max(0, x)$
- ☒ B) $1/(1+e^{-x})$ ✓
- ☐ C) $\tan(x)$
- ☐ D) x^2

✓ Which field heavily uses Deep Learning? *

1/1

- ☐ A) Agriculture only
- ☒ B) Image Recognition ✓
- ☐ C) Accounting only
- ☐ D) Civil Engineering only



✓ Autoencoders are primarily used for: *

1/1

- ☐ A) Classification
- ☒ B) Data compression and feature learning ✓
- ☐ C) Regression
- ☐ D) Sorting

✓ RBM consists of: *

1/1

- ☐ A) One input layer only
- ☒ B) Hidden and visible layers ✓
- ☐ C) Three hidden layers
- ☐ D) Only output layer

✓ Speech recognition is an application of: *

1/1

- ☐ A) Networking
- ☒ B) Deep Learning ✓
- ☐ C) Database Management
- ☐ D) Compiler Design



✓ Which is NOT an example of Representation Learning? *

1/1

- ☐ A) Autoencoder
- ☐ B) Deep Neural Network
- ☐ C) Principal Component Analysis
- ☒ D) Manual Feature Engineering



✓ The compressed representation in an Autoencoder is called: *

1/1

- ☐ A) Feature Map
- ☒ B) Bottleneck Layer
- ☐ C) Output Layer
- ☐ D) Activation Layer



✓ Medical image diagnosis commonly uses: *

1/1

- ☒ A) Deep Learning
- ☐ B) Bubble Sort
- ☐ C) Binary Trees
- ☐ D) Stacks



✓ Which of the following is an unsupervised deep learning model? * 1/1

- ☐ A) CNN
- ☒ B) Autoencoder ✓
- ☐ C) Logistic Regression
- ☐ D) SVM

✓ The width of a neural network refers to: * 1/1

- ☒ A) Number of neurons in a layer ✓
- ☐ B) Number of hidden layers
- ☐ C) Number of epochs
- ☐ D) Number of datasets

✓ A wider neural network mainly increases: * 1/1

- ☐ A) Layer count
- ☒ B) Number of neurons per layer ✓
- ☐ C) Number of datasets
- ☐ D) Number of outputs



✓ ReLU stands for: *

1/1

- ☐ A) Random Linear Unit
- ☒ B) Rectified Linear Unit ✓
- ☐ C) Reduced Learning Unit
- ☐ D) Recursive Learning Unit

✓ Leaky ReLU allows: *

1/1

- ☐ A) Positive values only
- ☒ B) Small negative values ✓
- ☐ C) Zero output always
- ☐ D) Outputs between 0 and 1

✓ RBM belongs to which type of graphical model? *

1/1

- ☐ A) Directed
- ☒ B) Undirected ✓
- ☐ C) Sequential
- ☐ D) Recursive



✓ Which of the following best describes Machine Learning? * 1/1

- ☐ A) A subset of Deep Learning
- ☒ B) A method where computers learn from data without explicit programming ✓
- ☐ C) A database management system
- ☐ D) A computer network protocol

✓ Which activation function helps solve the "dying ReLU" problem? * 1/1

- ☐ A) Sigmoid
- ☐ B) Tanh
- ☒ C) Leaky ReLU ✓
- ☐ D) Softmax

✓ Which company uses Deep Learning in recommendation systems? * 1/1

- ☒ A) Netflix ✓
- ☐ B) Microsoft Paint
- ☐ C) Notepad
- ☐ D) Calculator

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